

Olist E-Commerce Analytics

Customer Experience & Marketplace Performance

A visual and narrative report based on the supplied Brazilian e-commerce analysis notebook. Code cells and raw dataframe outputs are intentionally excluded; the report contains the analytical text, key metrics, and existing visualizations.

Dataset scope

Approximately 100,000 Olist marketplace orders covering September 2016 through October 2018, with information on orders, products, payments, sellers, customers, delivery, and reviews.

Analytical focus: marketplace performance over time; delivery and customer satisfaction; seller and geographic patterns; product categories; payment behavior; and root causes associated with low review scores.

1. Project Introduction

This project analyzes approximately 100,000 orders from the Olist Brazilian e-commerce marketplace between September 2016 and October 2018. The objective is to understand marketplace performance, customer satisfaction, delivery performance, seller and geographic patterns, product category performance, and payment behavior.

The analysis combines transactional evidence to identify patterns most strongly associated with customer satisfaction and dissatisfaction and translate those findings into practical business recommendations.

2. Business Problem

The central business challenge is to understand what is driving marketplace performance and customer satisfaction across orders, products, payments, sellers, customers, delivery information, and reviews.

3. Analytical Objectives

The analysis addresses six questions: how order volume, revenue, and review scores change over time; how delivery timing relates to reviews; how seller and customer geography relate to delivery, freight, and satisfaction; which categories generate the most orders and revenue; how payment methods and installments relate to order value and experience; and which factors are most strongly associated with low review scores.

Important analytical principle: the notebook explicitly treats the data as observational. Results therefore describe associations and patterns rather than definitive causal relationships.

4. Dataset Overview

The supplied notebook works with nine related datasets: orders, order items, order payments, order reviews, customers, products, sellers, geolocation, and product-category translation. The order-level analytical dataset brings these sources together for marketplace analysis.

Dataset scale

The notebook reports 99,441 orders, 112,650 order items, 103,886 payment records, 99,224 review records, 99,441 customers, 32,951 products, 3,095 sellers, more than one million geolocation records, and 71 category translations.

5. KPI Overview

99,441 — Total orders analyzed

R\$ 16,008,872.12 — Total payment value recorded

4.09 / 5 — Average customer review score

89.15% — On-time delivery rate

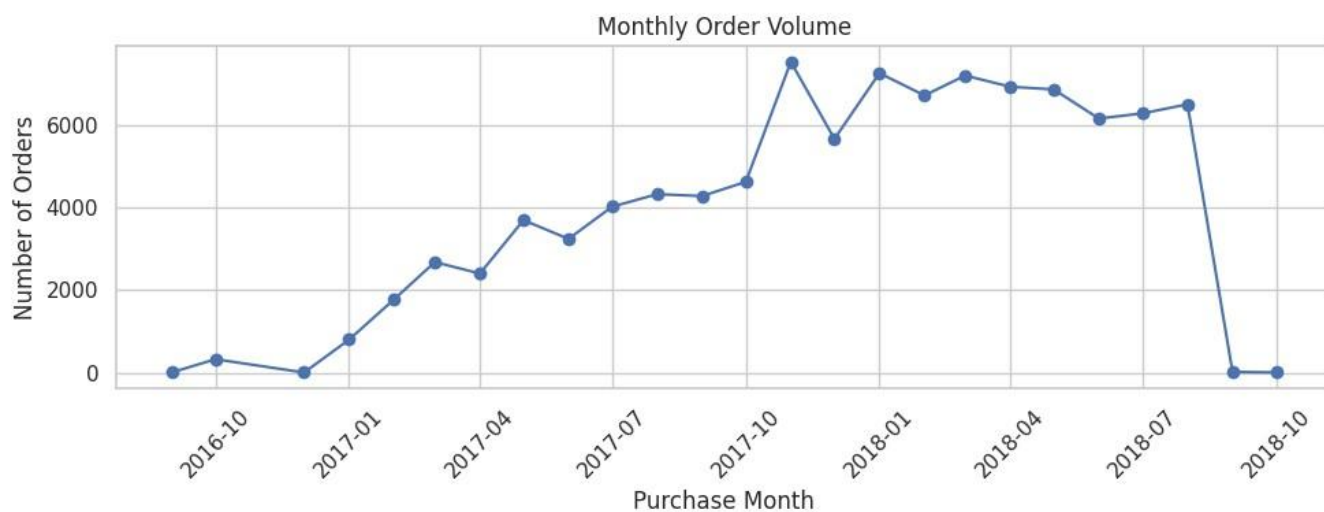
4.29 — Average review score for early/on-time orders

2.57 — Average review score for late orders

These headline metrics establish the overall marketplace scale and highlight the importance of delivery performance to customer experience.

6. Marketplace Performance Over Time

Order volume and revenue generally rise from 2017 into 2018, with the strongest activity occurring around late 2017 and early 2018. The final September and October 2018 observations are extremely small, so the sharp decline at the end of the charts should be interpreted as partial-period data rather than a clear collapse in marketplace activity.



Monthly order volume across the observed period.



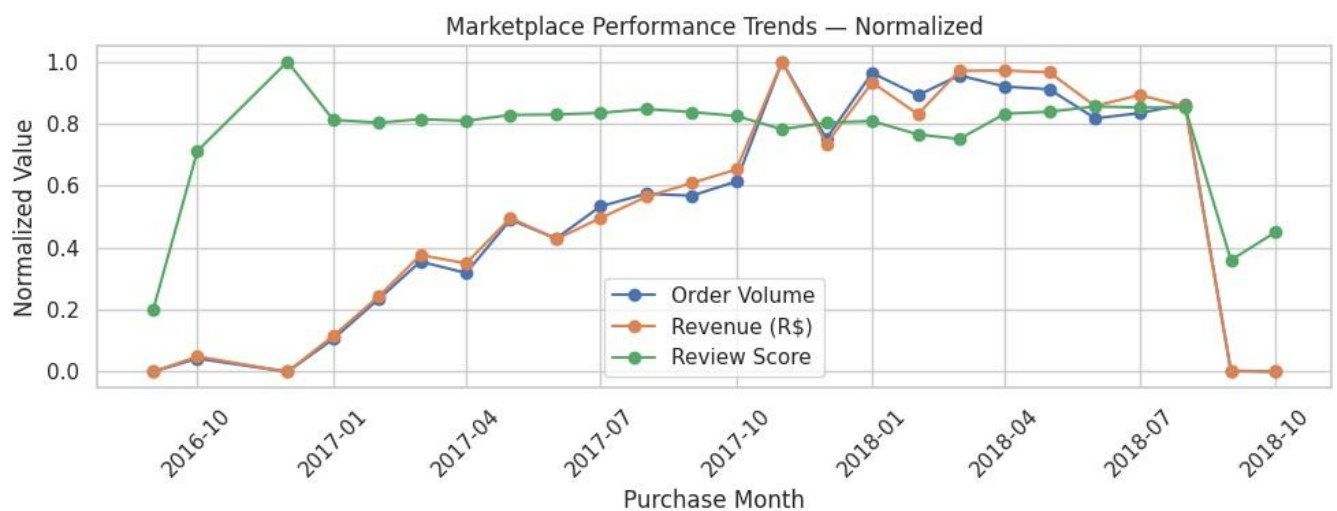
Monthly recorded revenue across the observed period.

Marketplace Performance — Customer Experience

Average review scores remain broadly around the 4-point level through most of the mature period, while the normalized comparison shows that order volume and revenue move closely together. Review performance is comparatively more stable than commercial volume.



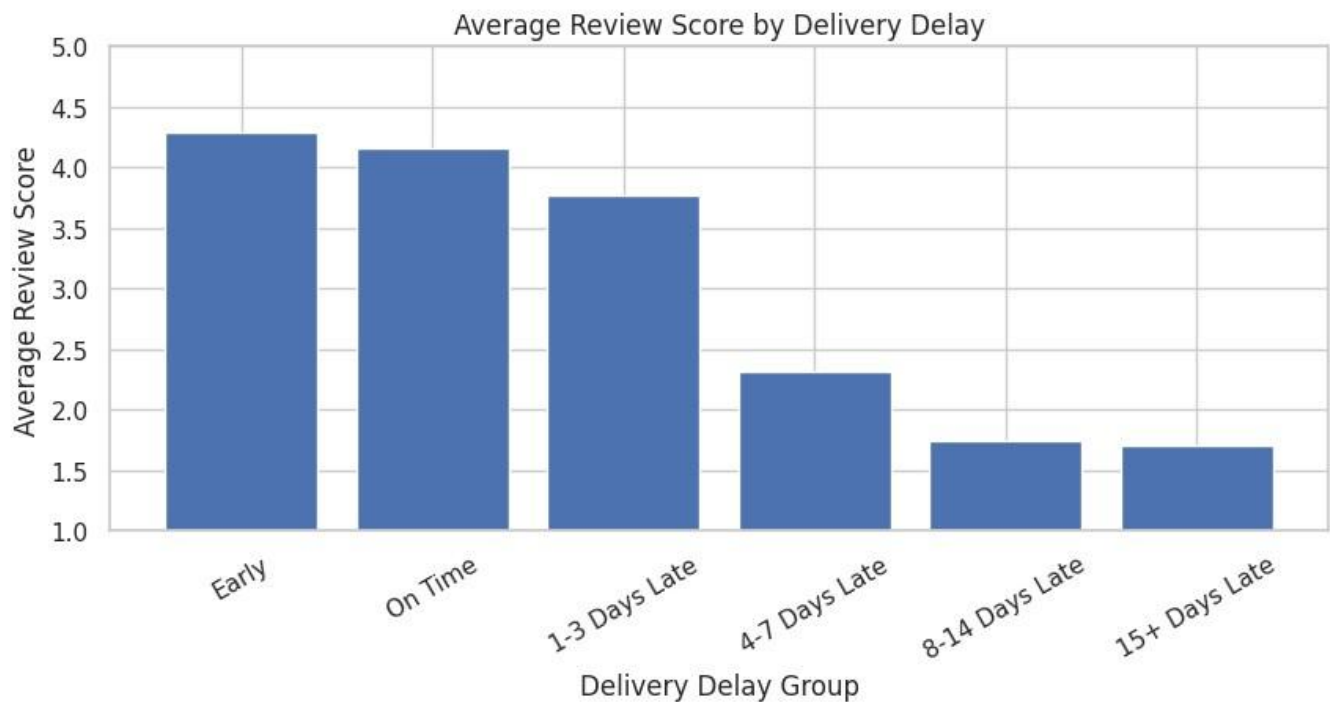
Monthly average customer review score.



Normalized comparison of order volume, revenue, and review score.

7. Delivery Performance & Customer Satisfaction

Delivery performance shows the clearest relationship with customer satisfaction. The notebook reports an average delivery time of 12.56 days, a median of 10.22 days, an on-time delivery rate of 89.15%, and an average review score of 4.09.



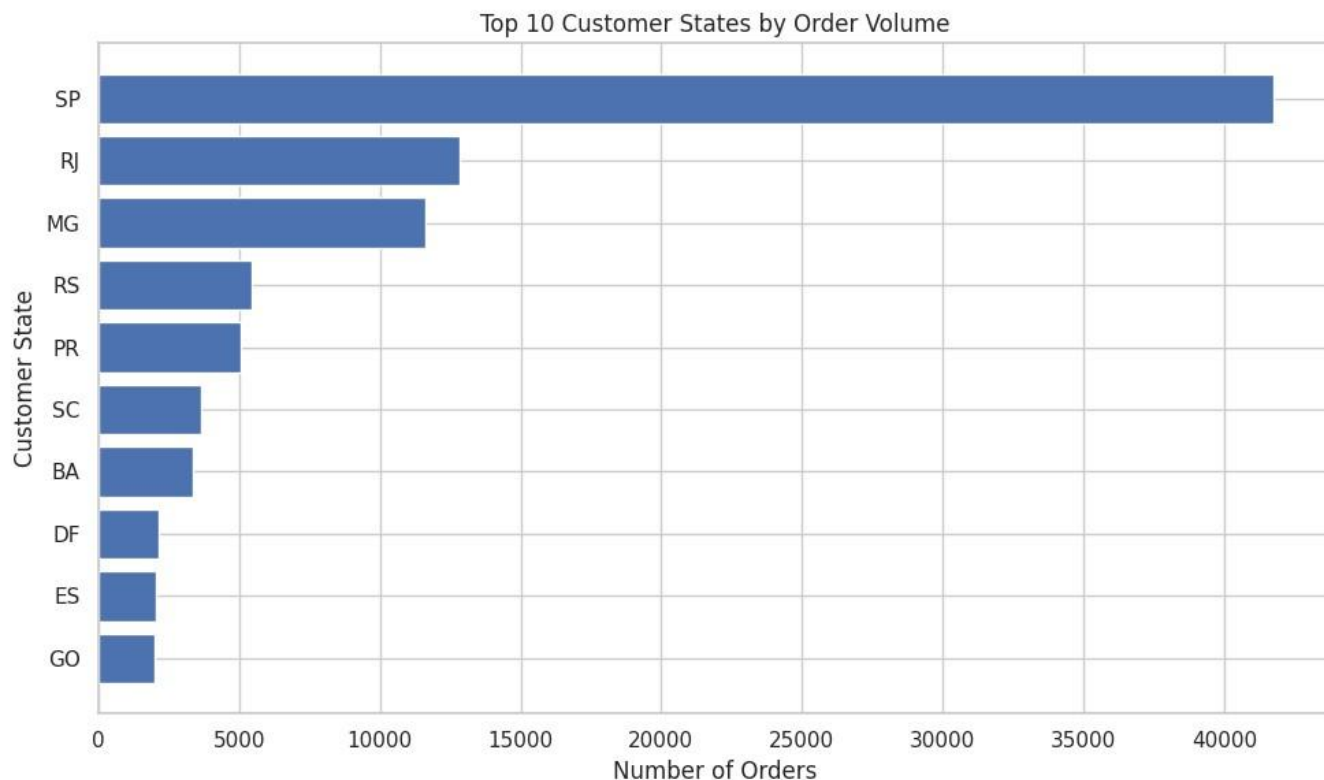
Average review score falls sharply as delivery delay increases.

Interpretation

Early deliveries receive an average review score of 4.30, compared with 4.16 for on-time deliveries. The score falls to 3.77 for orders 1–3 days late, 2.32 for orders 4–7 days late, 1.75 for orders 8–14 days late, and 1.71 for orders more than 15 days late. This pattern makes delivery reliability the strongest operational signal in the notebook's customer-experience analysis.

8. Seller & Geographic Patterns

Seller activity is highly concentrated geographically. São Paulo (SP) is the dominant seller state by revenue, followed by Paraná (PR), Minas Gerais (MG), Rio de Janeiro (RJ), and Santa Catarina (SC). On the customer side, SP also has the largest order volume by a wide margin.

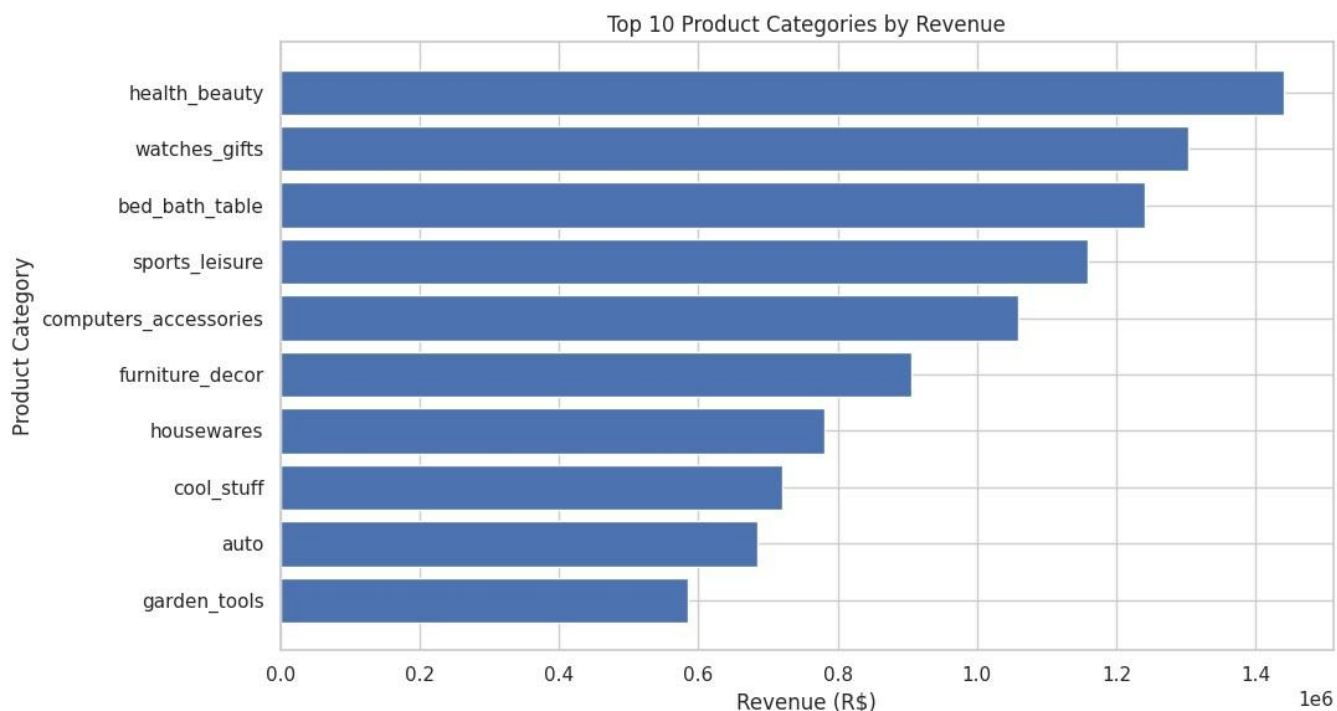


Top customer states by order volume.

The notebook reports a correlation of -0.089 between freight value and review score. This is a weak negative association: higher freight cost is related to slightly lower review scores in the analyzed data, but the relationship is not strong.

9. Product Category Performance

Health & beauty is the highest-revenue category in the notebook, while bed & bath table has the highest order volume. Other major categories include watches & gifts, sports & leisure, computers & accessories, and furniture & decor.



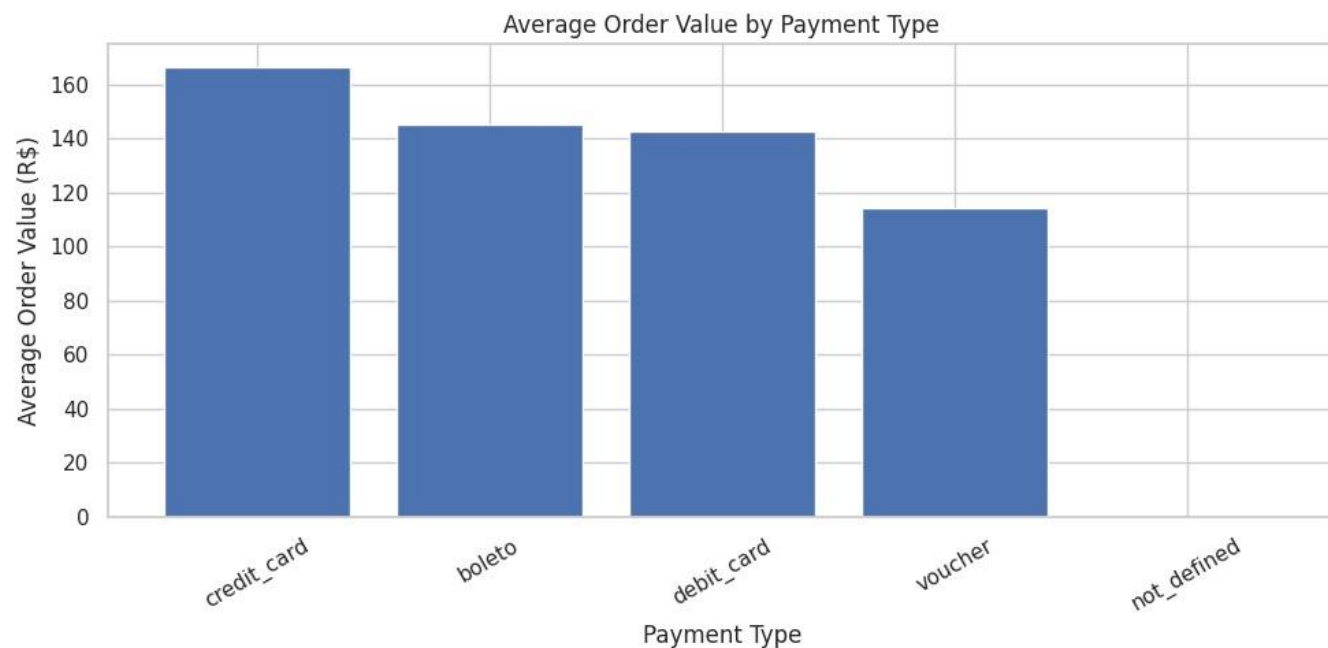
Top 10 product categories by recorded revenue.

For categories with at least 100 orders, office furniture has the lowest average review score at 3.62. It also has relatively high average freight of R\$53.95, making it an important category for operational investigation.

10. Payment Behavior

Credit cards dominate payment activity, with 76,504 orders and approximately R\$12.74 million in recorded revenue.

Boleto is the second-largest payment method by order count and revenue.



Average order value by payment type.

Average order value is highest for credit-card orders at about R\$166.47, followed by boleto at R\$145.03, debit card at R\$142.73, and voucher at R\$114.39. The notebook also reports only weak correlations between payment variables and review score, suggesting that payment behavior is much less strongly associated with satisfaction than delivery performance.

11. Root Cause Analysis of Low Review Scores

The notebook defines low reviews as scores of 1 or 2 and compares them with higher review scores. Low-review orders have an average review score of 1.22, an average delivery time of 20.22 days, and an on-time rate of 56.09%. Higher-review orders have an average review score of 4.58, an average delivery time of 11.44 days, and an on-time rate of 94.77%.



Distribution of customer review scores across analyzed orders.

Factors most strongly associated with review score

The notebook ranks factors by absolute correlation with review score. On-time delivery has the strongest positive correlation (0.446), followed by delivery time in days (-0.334) and delivery delay in days (-0.267). Order-item count (-0.116) and freight value (-0.089) are weaker, while payment value, order value, and installment count show very small correlations.

12. Key Findings

- Total orders analyzed: 99,441.
- Total payment value recorded: R\$16,008,872.12.
- Average customer review score: 4.09/5.
- On-time delivery rate: 89.15%.
- Average review score for early/on-time orders: 4.29.
- Average review score for late orders: 2.57.
- Highest-revenue product category: health & beauty.
- Highest-order-volume customer state: São Paulo (SP).
- Lowest-rated category with at least 100 orders: office furniture (3.62/5).

Overall interpretation: delivery performance is the dominant operational theme in the notebook. Late delivery is strongly associated with lower review scores, while payment-related variables show comparatively weak relationships with satisfaction.

All findings are observational associations and should not be interpreted as proof of causation.

13. Business Recommendations

1. **Prioritize late-delivery reduction.** Monitor orders approaching their estimated delivery date and intervene before they become late.
2. **Use delivery performance as a customer-experience KPI.** Segment reviews by delivery-delay group and investigate categories or regions with unusually poor results.
3. **Focus on high-volume and high-revenue areas first.** Operational improvements in major categories and regions can affect a larger share of marketplace activity.
4. **Monitor freight costs.** High freight costs combined with weak review scores should be investigated for possible logistics or customer-experience issues.
5. **Maintain payment flexibility.** Continue supporting multiple payment options while monitoring order value and customer-experience metrics across payment methods and installment levels.
6. **Build a recurring marketplace dashboard.** Track order volume, revenue, average review score, on-time delivery rate, delivery delay, freight value, and category performance.

14. Conclusion

The Olist dataset provides a detailed view of marketplace activity across orders, payments, products, sellers, customers, delivery performance, and reviews. The analysis combines these sources into an order-level analytical dataset and evaluates marketplace growth, delivery performance, geographic and seller patterns, category performance, payment behavior, and factors associated with customer satisfaction.

The strongest analytical emphasis is the relationship between delivery performance and customer reviews. Late delivery, delivery duration, freight cost, product category, geography, and payment characteristics can be compared to identify operational areas that deserve attention. Because the underlying data is observational, the results describe associations rather than definitive causal relationships.

The completed analysis therefore supports a practical marketplace-management approach: monitor performance over time, identify high-impact operational bottlenecks, prioritize high-volume areas for improvement, and continuously track customer satisfaction alongside logistics and commercial KPIs.

End of Report